QuantumTrace: Cryptographic Risk Matrix

Algorithm Classification Guide, Key Size Impact, and NIST PQC Standards

NIST FIPS 203 / 204 / 205 Standards

Shor's Algorithm

Impact: Complete Break. Solves integer factorization and discrete logarithms in polynomial time. Larger asymmetric keys (e.g., RSA-4096) provide no resistance.

Grover's Algorithm

Impact: Quadratic Speedup. Accelerates unstructured search spaces. Halves effective bit security (128-bit security effectively drops to 64 bits).

NIST PQC Standards

Impact: Official Compliance. Migration to ML-KEM (key encapsulation), ML-DSA, and SLH-DSA (quantum-resistant digital signatures).

Cryptographic Auditor Rule Table

Primitive	Key Size	Risk Tier	Quantum Threat	Effective Security	Recommended NIST Replacement
RSA	`1024 / 2048`	CRITICAL	Shor's Algorithm	0 bits (Broken)	**ML-DSA-65** (NIST FIPS 204)
RSA	`3072 / 4096`	CRITICAL	Shor's Algorithm	0 bits (Broken)	**ML-DSA-65** (NIST FIPS 204)
ECDSA / ECC	`secp256r1`	CRITICAL	Shor's Algorithm	0 bits (ECDLP broken)	**ML-DSA-44 / 65** (NIST FIPS 204)
ECDH / X25519	`256 / 384 bits`	CRITICAL	Shor's Algorithm	0 bits (Vulnerable KEM)	**ML-KEM-768** (NIST FIPS 203)
DSA / DH	`Any size`	CRITICAL	Shor's Algorithm	0 bits (Broken)	**ML-DSA** (FIPS 204) / **ML-KEM**
AES	`128 bits`	MEDIUM	Grover's Algorithm	**64 bits** (Insecure long-term)	**AES-256** (Increase key size)
AES	`192 bits`	MEDIUM	Grover's Algorithm	**96 bits** (Below 128-bit target)	**AES-256**
AES	`256 bits`	SAFE	Grover's Algorithm	**128 bits** (Grover-resistant)	**Compliant** (Keep AES-256)
SHA-256	`256 bits`	SAFE	Grover / BHT	128 bits (Collision resistance)	**Compliant** (or SHA-384/512)
SHA-1 / MD5	`160 / 128 bits`	CRITICAL	Classical + Quantum	0 bits (Classical collisions)	**SHA-256** / **SHA-3**
ML-KEM	`512/768/1024`	SAFE	PQC Resistant	NIST Security Level 1/3/5	**Native Standard** (NIST FIPS 203)
ML-DSA	`44/65/87`	SAFE	PQC Resistant	NIST Security Level 2/3/5	**Native Standard** (NIST FIPS 204)

Auditor Implementation Rule: For asymmetric algorithms (RSA, ECC, ECDH, DSA), any *key size* always yields a **Critical** severity under Shor's threat. For symmetric ciphers (AES), the *key size* is the deterministic factor: keys < 256 bits must be classified as **Medium**, whereas 256 bits is classified as **Safe**.